

Overview:

For this project, we will design, implement, test, and evaluate a 2D arcade-style game called “**Vault Escape**”. The game will revolve around a player controlling a robber who must navigate through a maze-like vault, collect rewards, and reach the exit within a limited time while avoiding various enemies and obstacles. The overall goal of the player is to collect all the basic rewards (money) before escaping through the vault’s exit within a limited time.

Game Concept:

In “**Vault Escape**”, the player controls the main character, a robber, using the WASD keys. The robber is motivated by greed, aiming to collect all the money (basic rewards) scattered around the vault. Additionally, diamonds (bonus rewards) occasionally appear for a limited time, offering extra points to boost the final score. The vault is heavily secured with enemies like guards, dogs, and static security cameras, making the journey a test of strategy and stealth.

Game Mechanics:

1. Maze-Like Vault:

- The vault is structured with walls and barriers, forming a complex maze. The player must navigate through this to reach the exit.

2. Player Control:

- The robber moves one cell at a time using WASD keys. The player must collect all the basic rewards (money) to unlock the exit.

3. Enemies:

- **All:** If a player is caught by any of the enemies, the player’s score is heavily penalized and may ultimately end the game once the player’s score reaches the negatives. This event also reduces the amount of time to clear the level, prompting more enemies to spawn.
- **Guards (Moving Enemies):** Patrol specific areas of the vault with a fixed movement pattern.
- **Dogs (Moving Enemies):** Patrols a specific area. Once a robber enters their proximity, the dogs will actively chase the robber until they are out of their proximity. Their proximity will expand as time goes on.
- **Cameras (Static Traps):** Stationary but dangerous. Catching the robber grants the movement speed guards and dogs based on frequency and duration of the robber caught by cameras.

4. Rewards:

- **Money (Basic Reward):** Scattered across the vault. The robber should collect all the money to unlock the exit and proceed to win the game.
- **Diamonds (Bonus Reward):** Randomly appear on the map for a short time, giving the player a chance to increase their score but not necessary for victory.

Win/Loss Conditions:

- **Winning:** The player wins by collecting all the money and successfully navigating to the vault’s exit before time runs out.
- **Losing:** The player loses if they fail to collect the basic rewards within the given time or if they are caught by the guards or dogs.

This game blends strategy, stealth, and quick reflexes, as the player must balance collecting rewards while avoiding detection and managing time. The tension builds as cameras increase enemy speed and proximity-based threats from dogs keep the player on edge.